

A RESEARCH PROGRAMME IN BOUNDED SELF-MONITORING

sīDUS

Towards Metarepresentational Ecology

DOCUMENT IV

E0 Protocol

Two-Stage Construct Validity

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specialist · quantity · way

Truth, maximally sought. Ambition tempered by humility.
Beauty tempered by goodness. Honesty above all else.

Sīdus: Towards Metarepresentational Ecology

IV. E0 Construct-Validity Protocol

Two-stage mock-up for validating internal-state monitoring before testing strict metarepresentation

Working methods companion - August 2026

Status: pre-experimental protocol. E0-A is specified strongly enough to guide implementation. E0-B is deliberately provisional and should be refined against the metarepresentation literature before confirmatory use.

***Purpose:** establish construct validity before introducing shared bottlenecks, multi-source competition, causal-topology claims, recursive allocator monitoring, or any ecology-level inference.*

0. Scope and decision rule

E0 is split into two independent stages because **internal-state monitoring is not automatically metarepresentation**.

- a) **E0-A - Monitoring substrate validity** asks whether a private internal state is genuinely hidden, incrementally represented, and causally used.
- b) **E0-B - Metarepresentational discrimination** asks whether a candidate higher-order signal represents a property or process of a first-order representation rather than merely tracking an internal state, public context, or another feedforward transformation.

The stages license different claims:

Stage passed	Claim licensed	Claim not licensed
E0-A	a causally used internal-state monitoring channel exists	the signal is a strict metarepresentation
E0-B	the tested architecture satisfies the selected operational criterion for property/process metarepresentation	metarepresentation is a natural kind, domain-general, conscious, or ecologically organized

E1 scarcity requires E0-A. Strict-metarepresentation experiments require E0-B. The broader programme should not infer metarepresentational ecology from E0-A alone.

Units. All rates, entropies and information quantities in this protocol are in **bits** ($\log_2$), per the standing conventions of Document I, §11A. Any implementation quantity produced in nats is converted at the point of reporting, not at the point of reading.

PART I - E0-A: MONITORING SUBSTRATE VALIDITY

1. Core architecture

Let the coordinator have ordinary public information

$$\mathcal{J}_t = \{o_{\leq t}, a_{< t}, h_{t-1}^U\},$$

where $o_{\leq t}$ is observation history, $a_{< t}$ is action history, and h_{t-1}^U is the coordinator's recurrent state before the special monitoring message arrives.

A specialist process has a private internal state

$$C_{v,t} \in \mathcal{C}_v.$$

A source-specific encoder produces an internal-state message

$$S_{v,t} = E_v(C_{v,t}, \mathcal{J}_t).$$

The coordinator/consumer U receives $S_{v,t}$ in addition to $\mathcal{J}_t$.

The intended causal chain is:

$$\mathcal{J}_t \parallel C_{v,t} \rightarrow S_{v,t} \rightarrow U.$$

E0-A asks three distinct questions:

$$\text{hiddenness} \neq \text{informativeness} \neq \text{causal use.}$$

2. Gate A1 - Residual hiddenness and potential value

2.1 Residual hiddenness

Define residual uncertainty:

$$u_v = \frac{H(C_{v,t} \mid \mathcal{J}_t)}{H(C_{v,t})}, \quad H(C_{v,t}) > 0.$$

The purpose is not to require independence between C_v and public history. It is to verify that a non-trivial informational problem remains after ordinary public information is used.

2.2 Practical estimator

The true conditional entropy is generally unavailable. Fit a calibrated held-out predictor $\hat{p}_0(c \mid \mathcal{J}_t)$ and estimate predictive cross-entropy:

$$\widehat{CE}_0 = \mathbb{E}_{\text{test}}[-\log_2 \hat{p}_0(C_{v,t} \mid \mathcal{J}_t)].$$

Because held-out cross-entropy upper-bounds the true conditional entropy by the predictor's approximation error, it should be treated as an **estimator/proxy**, not as exact entropy unless the decoder is demonstrably close to Bayes-optimal.

2.3 Direct-read headroom

Residual hiddenness alone is insufficient. Give the coordinator direct access to $C_{v,t}$ in an oracle condition and measure the gain over public information alone:

$$\Delta J_{\text{oracle},v} = J(\mathcal{J}_t, C_{v,t}) - J(\mathcal{J}_t).$$

A source can be hidden yet irrelevant. E0-A therefore requires both:

$$H(C_v \mid \mathcal{J}) > 0$$

and a preregistered minimally meaningful oracle benefit.

2.4 Gate A1

Pass only if:

1. residual hiddenness is reliably non-zero under the intended high-privacy condition;
2. the direct-read oracle shows that information about C_v can matter to the task;
3. public shortcut performance is quantified, not assumed absent.

Thresholds such as $u_v \geq 0.25$ should not be fixed by aesthetics. Calibrate pilot thresholds from task headroom and freeze them before confirmatory runs.

3. Gate A2 - Incremental transmission

3.1 Target quantity

Define the incremental information delivered by the monitoring message:

$$m_v = I(C_{v,t}; S_{v,t} \mid \mathcal{J}_t).$$

Equivalently,

$$m_v = H(C_{v,t} \mid \mathcal{J}_t) - H(C_{v,t} \mid \mathcal{J}_t, S_{v,t}).$$

3.2 Practical log-loss estimator

Fit two matched auxiliary predictors on held-out data:

$$\hat{p}_0(c \mid \mathcal{J}_t), \quad \hat{p}_1(c \mid \mathcal{J}_t, S_{v,t}).$$

Then estimate:

$$\hat{m}_v = \widehat{CE}_0 - \widehat{CE}_1.$$

This is an empirical conditional-information estimate under the chosen decoder class. Report cross-validation uncertainty and calibration diagnostics.

3.3 Rate spent

For a variational channel,

$$r_v = \mathbb{E} [D_{KL} (q(S_v | C_v, \mathcal{J}) \| p(S_v | \mathcal{J}))].$$

Under the intended variational construction,

$$I(C_v; S_v | \mathcal{J}) \leq r_v,$$

but finite-sample estimates need not obey the inequality exactly.

The central distinction remains:

$$r_v \neq m_v.$$

A channel may consume rate while adding little useful information.

3.3a Decomposition of the rate–information gap

[R2 · correction, August 2026 review pass]

The inequality $m_v \leq r_v$ was previously asserted as a property of “the intended variational construction.” It is in fact an identity plus a non-negative term, and writing it out converts a qualitative diagnostic into two estimable numbers.

For any q and p ,

$$\begin{aligned} \mathbb{E}_q [D_{KL}(q(S_v | C_v, \mathcal{J}) \| p(S_v | \mathcal{J}))] &= \underbrace{I(C_v; S_v | \mathcal{J})}_{m_v} \\ &\quad + \underbrace{\mathbb{E}_{\mathcal{J}} D_{KL}(q(S_v | \mathcal{J}) \| p(S_v | \mathcal{J}))}_{\Delta_v^{\text{prior}} \geq 0}. \end{aligned}$$

So

$$r_v = m_v + \Delta_v^{\text{prior}}, \quad \Delta_v^{\text{prior}} := \mathbb{E}_{\mathcal{J}} D_{KL}(q(S_v | \mathcal{J}) \| p(S_v | \mathcal{J})) \geq 0.$$

[Revised after release-candidate review.] The mismatch term is an expectation over $\mathcal{J}$, not a bare divergence. $D_{KL}(q(S_v | \mathcal{J}) \| p(S_v | \mathcal{J}))$ is a function of the random variable $\mathcal{J}$; only its expectation is the scalar that completes the identity. Because this is stated as an identity it has to be exact, and an implementation that omits the average would compute a per-timestep quantity and mis-report it.

Three things follow.

1. $r_v \geq m_v$ **is robust, not an assumption.** It holds for any prior, learned or fixed. Finite-sample estimates may still violate it; the population statement does not depend on the encoder having been trained well.
2. **The Gate A4 signature decomposes exactly.** The pattern $m_v \approx 0$, $r_v > 0$ —recorded there as “possible redundant coding, inefficient optimization, or target-metric mismatch”—separates cleanly: the rate is spent on Δ_v^{prior} , i.e. mismatch between the aggregate posterior $q(S_v | \mathcal{J})$ and the prior, rather than on information about C_v . Estimate Δ_v^{prior} directly and report it; do not infer it by subtraction alone.

3. **It answers the “soft penalty with extra steps” objection.** With a *learned* prior fitted to the aggregate posterior, Δ_v^{prior} shrinks and the variational surrogate tightens toward the true conditional information. The surrogate’s looseness is therefore a measurable quantity rather than an unknown, which is what licenses using it as a rate meter at all. It does **not**, on its own, establish that a trained run satisfies the shared constraint —see Arm S’s realised-violation feasibility diagnostic in Document II, §10. The alphabet arm there is optional currency-sensitivity work, not a requirement for E1.

3.4 Gate A2

Pass only if, under a non-binding communication condition:

1. $\hat{m}_v$ is meaningfully above zero with uncertainty reported;
 2. $r_v > 0$ and the encoder has not collapsed;
 3. performance approaches the direct-read ceiling enough to show that the channel can solve the intended information problem.
-

4. Gate A3 - Selective causal use

Decodability is not use.

4.1 On-support counterfactual scrambling

Replace the actual message with a valid message associated with an alternative reachable controller state:

$$S_{v,t} \leftarrow S'_{v,t} \sim q(S_v \mid C'_{v,t}, \mathcal{I}_t), \quad C'_{v,t} \neq C_{v,t}.$$

The intervention should remain near trained support. Avoid arbitrary zeroing unless zero is itself a trained message state.

4.2 Primary outcome

Measure the **proximal decision or policy variable** predicted to depend on C_v , not only total episode return.

Examples:

$$\Delta\pi_v, \quad \Delta P(a_t \mid \cdot), \quad \Delta J_v.$$

Total return J is secondary when downstream compensation could mask a genuine local causal effect.

4.3 Non-target preservation

At the intervention time or within a preregistered proximal window, verify that non-target quantities remain approximately intact:

$$C_u, \quad S_u, \quad m_u, \quad u \neq v.$$

Later downstream changes caused by the intervention are legitimate causal consequences and should not be misclassified as intervention leakage.

4.4 Gate A3

Pass only if:

1. the on-support message intervention causes the predicted local downstream change;
 2. the effect exceeds a preregistered minimally meaningful threshold, not merely a significance threshold;
 3. first-order/private target execution is preserved at intervention time;
 4. non-target channels are preserved over the specified proximal window.
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5. Gate A4 - Shortcut sensitivity

E0-A should demonstrate that the special monitoring channel is valuable **because information is missing**, not merely because the architecture has learned to use an arbitrary extra input.

Construct legitimate environment conditions that vary the public inferability of C_v :

$$H(C_v \mid \mathcal{I})_{\text{high privacy}} > H(C_v \mid \mathcal{I})_{\text{low privacy}}.$$

As public observability increases, predict:

$$H(C_v \mid \mathcal{I}) \downarrow \Rightarrow m_v \downarrow$$

and

$$\text{causal benefit}(S_v) \downarrow.$$

Do **not** require $r_v \rightarrow 0$ in the absence of a channel cost. Redundant coding can persist even when incremental information vanishes.

If an explicit message cost is imposed, then efficient learning additionally predicts:

$$r_v \downarrow.$$

The pattern

$$m_v \approx 0, \quad r_v > 0$$

should be recorded as possible redundant coding, inefficient optimization, or target-metric mismatch rather than defined away.

Gate A4

Pass only if the informational and causal value of S_v changes in the predicted direction as public observability is manipulated within the task distribution. **[Second external review pass.]** The manipulation check is a confirmatory test, not an impression of direction: decline thresholds—including any ratio-of-decline criterion—are preregistered before confirmatory runs, and the comparison is made with reported confidence intervals on $\hat{m}_v$ and on the causal-benefit estimate, never with bare ratios of point estimates.

5A. Non-gating diagnostic - cross-source leakage must be measured conditionally

[R3 ·correction, August 2026 review pass —preventive]

No cross-source independence check appears in the protocol above. Earlier working sketches used an unconditional mutual information $I(S_u; S_v)$ as a “cheating” diagnostic, and any future implementation is likely to reach for the same quantity. It is the wrong one, and the error is silent.

Sources that share public context will exhibit $I(S_u; S_v) > 0$ **legitimately**, because both messages are conditioned on $\mathcal{J}_t$. An unconditional test therefore flags correct architectures as leaking. The right quantity conditions on the public information the consumer already holds:

$$I(S_u; S_v \mid \mathcal{J}_t),$$

and for $K > 2$ sources, the total correlation given $\mathcal{J}$:

$$TC(S_1, \dots, S_K \mid \mathcal{J}) = \sum_v H(S_v \mid \mathcal{J}) - H(S_1, \dots, S_K \mid \mathcal{J}).$$

Non-zero conditional dependence is still not automatically a fault: encoders may legitimately coordinate under a shared budget.

[Revised after release-candidate review.] An earlier version named the fault condition as $I(C_u; S_v \mid \mathcal{J}) > 0$. That is still too strong. If the sources are conditionally dependent,

$$C_u \not\perp C_v \mid \mathcal{J},$$

then $C_u \rightarrow C_v \rightarrow S_v$ forms a chain given $\mathcal{J}$, so S_v carries information about C_u **indirectly and legitimately**, even though encoder v never saw C_u . By data processing, $I(C_u; S_v \mid \mathcal{J}) \leq I(C_u; C_v \mid \mathcal{J})$, which is positive whenever the sources are dependent. Flagging that as leakage would condemn a correct architecture.

The diagnostic that isolates unauthorised access conditions on the source the encoder is entitled to see:

$$I(C_u; S_v \mid C_v, \mathcal{J}), \quad u \neq v.$$

Under the intended factorised architecture $S_v \sim q_v(S_v \mid C_v, \mathcal{J})$, this quantity vanishes exactly, absent shared private randomness or another forbidden path. Non-zero values indicate that encoder v has access it should not have—which is the quantity that would break the per-source attribution on which Gate A3, Gate A4 and hypothesis H3 all depend.

Report $I(C_u; C_v \mid \mathcal{J})$ alongside it. Source dependence sets how much legitimate cross-decodability to expect, and without it the conditional test has no reference scale.

Finally, prefer architecture to inference. The access restriction should be **enforced in code**—encoder v is wired to receive only C_v and $\mathcal{J}$ —and these information quantities used as **manipulation checks on that wiring**, not as a method for inferring architecture from mutual information alone.

6. E0-A pass/fail rule

E0-A licenses E1 scarcity only when all four gates pass across independent seeds. The cross-source leakage diagnostic in §5A is recommended and reportable when relevant, but it does **not** constitute a fifth gate:

Gate	Required evidence
A1 - Residual hiddenness + headroom	public information leaves non-trivial uncertainty and direct-read information can improve relevant behavior
A2 - Incremental transmission	non-binding channel transmits additional target information and avoids collapse
A3 - Selective causal use	on-support message intervention produces the predicted local downstream effect with proximal non-target preservation
A4 - Shortcut sensitivity	increasing legitimate public observability reduces incremental message information and its causal value

If a gate fails, do not proceed to multi-source scarcity as if the construct were intact. Re-engineer the task, simplify the channel, or revise the claim.

PART II - E0-B: PROVISIONAL METAREPRESENTATIONAL DISCRIMINATION

7. Scope of E0-B

E0-B is deliberately less mature than E0-A.

Its purpose is not to settle the philosophy of metarepresentation. It asks whether a synthetic experiment can distinguish a candidate **property/process metarepresentation** from simpler alternatives under explicitly stated operational criteria.

[Second external review pass —design commitment executed.] E0-B is not run under a single chosen definition. The three criteria of Document I §2.3 disagree in principle, so the protocol is **definition-indexed**:

$$E0-B^{\text{Shea}}, \quad E0-B^{\text{RR}}, \quad E0-B^{\text{Process}},$$

run over the same architectures, preregistered manipulations and evidence tiers. A system may satisfy one operationalization and fail another. That outcome is not experimental failure. If the criteria repeatedly carve systems differently, the warranted conclusion is that “**metarepresentation**” **does not yet behave as one empirically unified construct**, and the programme should then speak specifically of representational-property monitoring, process representation, or representational redescription —whichever actually survived —rather than of metarepresentation simpliciter.

E0-B should be refined against Proust, Shea, Cleeremans/SOMA, and process-metarepresentation work before confirmatory use.

8. Minimal architecture

Let an external latent state X_t generate observation O_t . A first-order system produces:

$$R_t = F_{\eta_t}(O_t),$$

where η_t controls a hidden property of the representational process, for example:

- noise/reliability regime;
- systematic bias;
- source type;
- gain/transformation rule;
- mapping family.

Construct at least three comparison systems:

B0 - First-order/public shortcut

Predicts the task target or η_t from public context, observation, or ordinary downstream activity without an explicit model of a representational property/process.

B1 - Internal-state monitor

Represents a private controller/process state C_t , but is not required to represent a property of R_t or the process F_{η_t} .

B2 - Candidate property/process metarepresentation

Produces a state M_t whose task-relevant target is explicitly:

$$\phi(R_t)$$

or

$$\eta_t,$$

where $\phi(R_t)$ may be reliability, source, correctness probability, or another preregistered representational property.

9. Minimum discriminators

E0-B should not pass merely because M_t decodes η_t .

At minimum, the design should include matched conditions that:

1. hold external target X_t fixed while η_t changes;
2. hold first-order choice/output fixed while reliability/source/process regime changes where feasible;
3. include cases in which M_t can misrepresent its target;
4. compare against a public-context shortcut with matched capacity;
5. perturb the target property/process and test whether M_t changes as predicted;
6. perturb M_t while preserving the first-order representation and test whether its intended consumer changes appropriately.

A provisional evidence hierarchy is:

$$\begin{aligned} \text{decodable} &< \text{conditionally specific} < \text{misrepresentable} \\ &< \text{target-dependent} < \text{causally used.} \end{aligned}$$

10. Provisional E0-B pass criterion

E0-B passes **for the selected operational definition only** if the candidate B2 architecture:

1. carries held-out information about the preregistered representational property/process after public and first-order shortcut predictors are controlled;
2. can dissociate from the first-order output and can misrepresent its meta-target;
3. responds selectively to intervention on the target representational property/process;
4. causally affects the intended downstream consumer when manipulated while first-order representation is preserved;
5. outperforms simpler B0/B1 explanations under held-out interventions, not merely under in-distribution decoding.

Outcomes are reported **per definition-index** as a vector of pass/fail results across the three operationalizations, not collapsed into a single verdict; a split outcome is a substantive finding about the construct, per §7.

If these conditions cannot be met under any index, the programme should not use the phrase **strict metarepresentation** for that system.

E0-B does **not** establish that this operationalization exhausts the concept of metarepresentation.

PART III - ASSUMPTIONS AND LIMITATIONS

11. Assumptions

1. **Ground truth is available by construction.** The synthetic environment gives the experimenter access to C_t , R_t , η_t , and intervention targets that biological work would not provide directly.
2. **Candidate variables are supplied by the experimenter.** E0 tests representation and use of known targets; it does not test spontaneous discovery of an internal ontology.
3. **Auxiliary decoders are adequate enough to support conditional-information estimates.** Their limitations must be measured through held-out calibration and capacity checks.
4. **On-support interventions can be designed without destroying unrelated computations.** This must be empirically verified rather than assumed.
5. **Task value is a legitimate local criterion.** Causal use is defined relative to the tested task and consumer; absence of use in one task does not prove universal absence of function.
6. **Representational properties/processes are operationally specified before data inspection.** E0-B cannot infer a meta-target after the fact from whichever latent dimension decodes best.

12. Limitations

12.1 E0-A validates monitoring, not metarepresentation

A perfectly successful E0-A system may still be only a sophisticated internal telemetry channel.

12.2 E0-B is definition-relative

Metarepresentation is contested. E0-B can validate one operational criterion against specified alternatives; it cannot settle all philosophical or cognitive-scientific definitions.

12.3 Conditional information estimation is model-dependent

Cross-validated log-loss differences approximate conditional information only relative to the decoder class and data regime. Poor auxiliary predictors can create misleading residual uncertainty or information estimates.

12.4 Synthetic modularity is an experimental scaffold

Clean private states, encoders, and intervention points are engineered conveniences. They should not be read as claims about anatomical modules or a literal neural information seam.

12.5 Causal use is consumer-relative

A message may be unused by the tested coordinator yet useful elsewhere. E0 establishes only the causal relation explicitly tested.

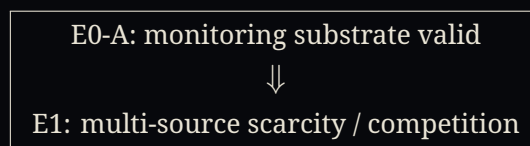
12.6 Redundant coding is possible

$m_v \rightarrow 0$ does not imply $r_v \rightarrow 0$. Efficient rate reduction requires an actual cost or opportunity pressure.

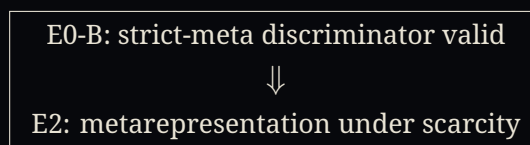
12.7 Local validity does not establish ecology

Even successful E0-A and E0-B experiments validate only local constructs. Ecology-level claims require reproducible interactions among multiple independently validated systems across tasks or architectures.

13. Pipeline after E0



and separately:



Passing E0-A does not imply E0-B. Passing either does not imply ecology.

E0 exists to prevent later sophistication from compensating for a bad construct. If the substrate cannot survive these tests, stop before building the ecology around it.

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